ArtGuard

ASTANA IT UNIVERSITY

Risk assessment, prototype and investment pitch

Introduction to Finance
Astana IT University · Team 03 · SE-2533
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GitHub: https://github.com/ONI-COLLI/Artguard
University OneDrive video: upload and institutional sharing pending

Educational simulation. Fictional customers, planning figures and test ETH only.

The licensing problem

Creators need clear evidence of the work and the agreed usage terms.

Buyers need a licence and a reliable delivery process.

Co-authors need a traceable record of their agreed shares.

The first proposed market is visual creators and small-business buyers in Kazakhstan. Demand and willingness to pay still need validation.

Sources: Ciriello et al. (2023); Frattolillo (2024).

Creator plans and service revenue

Table 1. Monthly creator plans

Plan	Price per month	Sales commission
Free	$0	10%
Premium	$10	5%
Business	$50	0%

Promotion: $5 for seven days. Base forecast assumes no promotion revenue until demand is validated.

Source: implemented plan rules. USD prices are reference prices; prototype payments use test ETH.

The implemented local architecture

React and wallet

Forms, signatures
and per-tab sessions

→

Express and SQLite

Accounts, quotes
and protected files

→

Anvil EVM

Hashes, escrow
and wallet licences

Files stay off-chain. The local operator controls Anvil. Built-in demo wallets use the server’s restricted custodial signer.

Figure 1. Actual application and blockchain boundaries

Source: current repository. Foundry (n.d.); OpenZeppelin (n.d.).

Escrow separates payment from acceptance

1 Register

Creator records file and terms fingerprints.

2 Deposit

Buyer locks the exact quoted test ETH amount.

3 Deliver

Seller confirms within 72 hours.

4 Review

Buyer has 24 hours to accept or dispute.

5 Release

Acceptance starts the licence and credits shares.

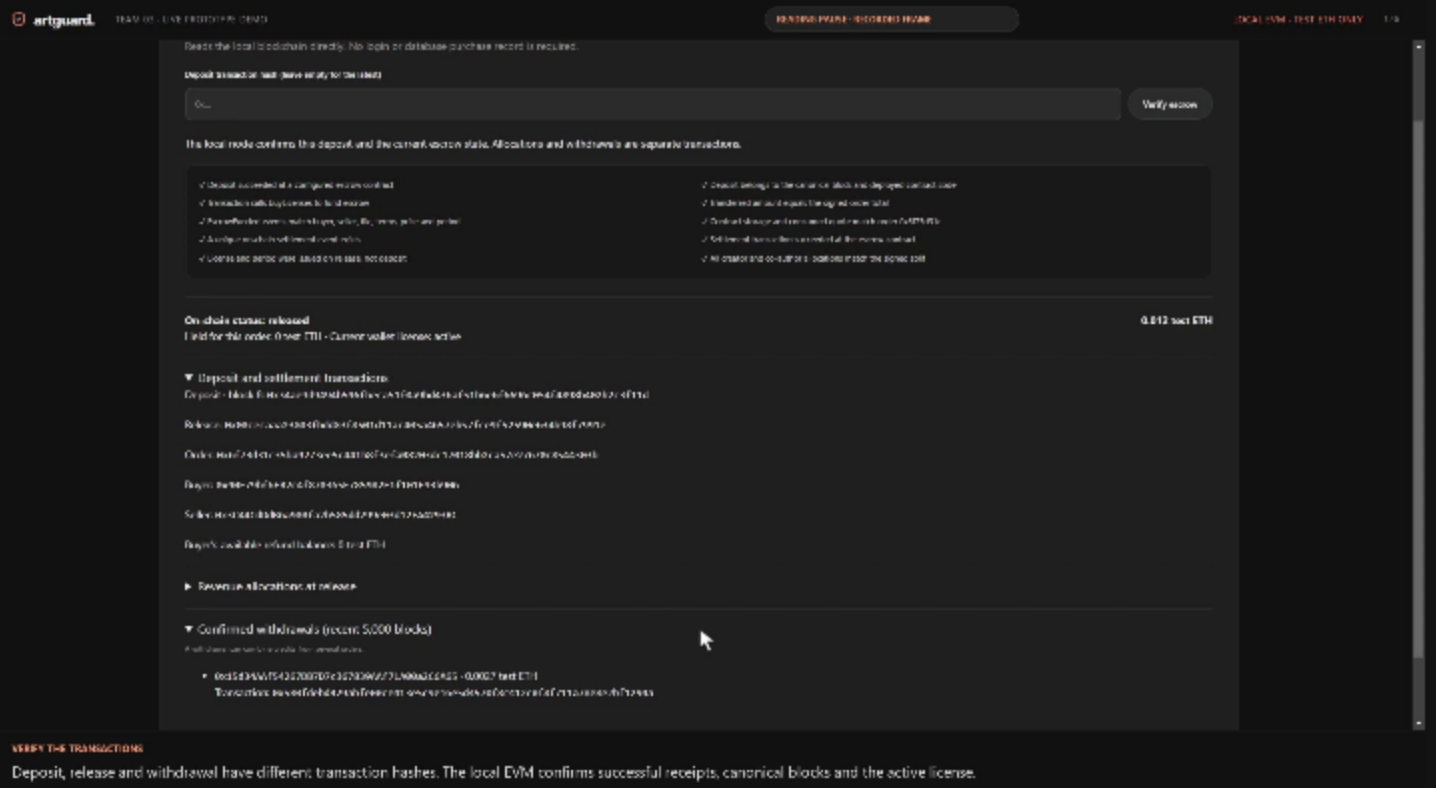
6 Withdraw

Each recipient withdraws in a separate transaction.

Figure 2. Current escrow sequence. A dispute blocks ordinary release; unresolved disputes permit refund after seven days.

Source: EscrowLicenseMarketplace.sol. All amounts use local test ETH.

Recorded deployment and purchase



Play Team03_FP_Demo.mp4

3:00 exactly
English on-screen explanation
Deployment and local receipts
Escrow, licence and payout
Actual tests and passing results

Figure 3. Actual local recording. Source: saved defence run ccab5e4f-561c-473b-b35c-44019c1d437c.

Security controls and residual trust

Table 2. Controls that actually exist

Threat	Implemented control	Remaining risk
Reentrancy	Clear credit before transfer; nonReentrant	Independent audit remains planned
Tampered or replayed quote	EIP-712, buyer, expiry and quote ID checks	Trusted signer can authorise bad terms
Unauthorised settlement	Wallet and state-transition checks	Arbiter compromise affects disputes

No implemented multisig, emergency pause or signer rotation is claimed. The future audit budget is an assumption.

Sources: current contracts; Kushwaha et al. (2022a, 2022b); OpenZeppelin (n.d.).

Market alternatives and the proposed difference

Table 3. Competitor categories

Category	Existing focus	ArtGuard question
Stock licensing	Content discovery and usage terms	Does an escrow workflow add useful assurance?
Infringement services	Find and address unauthorised use	Can buyers and creators use licence evidence more easily?
NFT marketplaces	Token ownership and trading	Can a wallet-bound licence work without issuing a token?

These services are competitors or reference categories, not clients or partners. No adoption or market-share claim is made.

Sources: Adobe (n.d.); Pixsy (n.d.); Bhujel and Rahulamathavan (2022).

The main risks affect the investment decision

Table 4. Highest-priority risks

Risk	Score	Response
Weak demand	20	Validate retention and willingness to pay
Funding shortfall	20	Stage commitments and rerun downside cases
Signer or arbiter compromise	15	Independent review and recovery design
Regulatory classification	15	Model-specific Kazakhstan legal review
False authorship and delivery disputes	12	Clear evidence, takedown and appeal process

Legal pages demonstrate a workflow. Commercial activity needs a model-specific review under Kazakhstan law and any relevant AIFC rules.

Source: own risk matrix. Legal scope: Law No. 193-VII (2023), Law No. 94-V (2013), AIFC (n.d.).

The forecast uses explicit assumptions

Table 5. Monthly demand anchors

Metric	Month 12	Month 24	Month 36
Licence sales	400	1,300	2,500
Premium creators	150	350	600
Business creators	20	60	120

Average licence: $30. Sales-value mix: 50% Free, 30% Premium, 20% Business. Weighted commission: 6.5%.

Processing: 2.5% of gross buyer payments plus subscriptions. Frozen planning FX: 447.94 KZT/USD, not a current-rate claim.

Source: scripts/finance-model.mjs. All targets are hypothetical and require team review.

Revenue growth does not repay all build losses

Table 6. Three-year forecast

Metric	Year 1	Year 2	Year 3
Platform revenue	6.50	34.39	71.39
Cash operating costs	29.17	44.26	65.50
Operating free cash flow	-22.67	-9.88	5.90
Cash after financing	17.33	7.46	13.35

Gross licence volume is separate from platform revenue. Creator proceeds and locked escrow funds are not company profit.

Source: executable monthly model. Units: KZT million. Rounded values may not sum exactly.

The three-year investment case remains negative

KZT -21.43m

Three-year project NPV

First positive monthly EBITDA: Month 23.
Payback: not reached by Month 36.
NPV including a 5% terminal-growth case: KZT -5.58m.

A funded cash balance does not establish an attractive project return.

Source: own calculations at 25% annual discount rate. Development costs are already in Year 1 FCF.

Downside scenarios require more funding

Table 7. Sensitivity of value and funding

Metric	Base	A	B	C
NPV Years 1–3	-21.43	-37.91	-38.47	-54.41
Peak cash need	35.76	55.70	56.60	82.47
Extra above KZT 40m	0.00	15.70	16.60	42.47
Payback by Month 36	No	No	No	No

Reduced scope or delayed hiring needs a new forecast. Additional downside capital can dilute the proposed 26% stake.

Source: own model. A: customers −30%; B: operating costs +20%; C: combined. Units: KZT million.

Funding terms need evidence and repricing

Table 8. Proposal retained for critical evaluation

Term	Illustration
Funding	KZT 25m in Month 1 + KZT 15m in Month 10
Equity	26% after both tranches
Implied post-money value	KZT 153.85m
Base peak cash need	KZT 35.76m
Illustrative investor return	1.56× MOIC; 17.7% annual IRR

The assumed exit multiple is unverified. The illustrative IRR falls below the 25% hurdle. A staged pilot is the recommended next decision.

Source: own illustrative terms and cash-flow calculation. No real investment is solicited.

The team must own and explain the work

Table 9. Review responsibilities

Member	Role	Area
Assan Dairabaev	CEO	Introduction, pitch and conclusion
Demezhan Kaidar	CTO	Architecture, contracts, security and technical evidence
Yevgeniy Tyo	CFO	Financial assumptions, model and sensitivity analysis
Kalybek Zhumabekov	COO	Operations, risk and regulatory analysis

Next: validate demand, review security and update the funding decision.

Each member provides their own commits and self-evaluation.

Source: roles confirmed by the project representative. Actual contribution remains individually documented.

Conclusion

The local prototype demonstrates registration, escrow, a wallet-bound licence and a real co-author withdrawal.

The commercial case remains conditional. The current three-year NPV is negative and downside scenarios exceed the proposed funding.

A staged validation pilot, independent review and revised economics should precede a hypothetical investment.

The course uses fictional data and test assets only.

Source: prototype, recorded local evidence and the revised financial model.

AI Disclosure Statement

AI assisted code, contracts, tests, interface explanations, demonstration tooling, English captions, report editing, references, financial-model code and presentation preparation.

This is substantial assistance. The team must review the design, assumptions, calculations and conclusions, and every member must explain the code independently.

No personal contribution, Scopus login, customer survey, coverage percentage or completed external audit is invented.

Tool and exact model supplied by the project representative: Codex, GPT-6.1 Sol.

Responsible student: Assan Dairabaev

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